BIOMETRIC BLOOD GROUP IDENTIFICATION SYSTEM

Final Year B.E. Major Project • Deep Learning AI Analysis Report

MEDICAL DISCLAIMER: AI-based preliminary estimation only. This result must be confirmed using a standard laboratory blood typing test before medical use. This system does not replace standard serological blood typing.

1. EXAMINATION & SAMPLE AUDIT DETAILS

Report ID:	BG-000182	**Timestamp:**	2026-09-05 12:38:00
Input Source:	Mantra MFS100 Scanner	**Evaluation Status:**	**NORMAL**
Model Name:	ResNet50 Transfer Learning	**Confidence:**	**62.79%**

2. BIOMETRIC IMPRESSION & PREDICTED BLOOD GROUP



Predicted Blood Group:

A-

Confidence Score: **62.79%**

Threshold: 60.0%

Classification: High Confidence

Derived ABO Group: **A** (62.9% prob) | Rh Factor: **-**

Rh FACTOR UNCERTAINTY ADVISORY: Rh factor (±) determination is uncertain (moderate 8-class confidence (62.8%); predicted Rh-negative requires serological validation due to weak ridge-Rh correlation). ABO group appears to be 'A' (62.9% confidence). Standard serological laboratory blood typing is required to confirm Rh factor.

3. 8-CLASS BLOOD GROUP PROBABILITY DISTRIBUTION

A+	A-	AB+	AB-	B+	B-	O+	O-
0.1%	62.8%	0.0%	2.8%	0.3%	32.0%	1.8%	0.2%